A dimension bound of ten for right-angled hyperbolic polytopes

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Abstract

We prove that a finite-volume right-angled hyperbolic polytope has dimension at most ten. The new argument excludes the mixed case in dimension eleven. Canonical three-faces define an undirected opposition graph on finite vertices, with one invariant Lorentz vector and one fixed null direction on each component. A structured positive-part inequality, applied only to directions whose edges end ideally, gives $2B \le 84f + 3E/2$. Here B counts the canonical three-faces, f the finite vertices, and E a nonnegative three-face excess. Polygon incidences, a one-cusp exclusion and a balanced corner count give the incompatible strict inequality $3722B > 159195f + 2794E$. The geometric and counting arguments are given in full. The scalar proof uses exact moment identities and finite rational positivity certificates for seven, eight and ten directions. Published bounds for the compact and entirely ideal cases, together with face heredity, give the universal statement. A separate conditional Lean module checks the final count logic; it does not formalize the hyperbolic geometry. No new example or sharpness assertion is made.

1 Statement and conventions

Theorem 1.1. *Every finite-volume right-angled hyperbolic polytope has dimension at most ten.*

We will exclude a polytope having both finite and ideal vertices in $\mathbb{H}^{11}$. Any higher-dimensional finite-volume right-angled polytope has an eleven-dimensional face. Faces of dimension at least three inherit right angles and finite volume: their finite and ideal links are faces of orthant and compact cube links. The eleven-face cannot be compact or entirely ideal, by the published upper bounds four and six [1, 2]. The exclusion therefore proves Theorem 1.1 without using a previous unrestricted dimensional bound.

Let $P \subset \mathbb{H}^{11}$ be a hypothetical mixed polytope. Write $f > 0$ for its finite vertices, $c > 0$ for its ideal vertices, and a_j for its j-faces. We use Lorentz signature $(11, 1)$, future unit points $q^2 = -1$, and outward unit normals n with wall inequality $\langle x, n\rangle \le 0$. Distinct outward normals have nonpositive products. At a finite vertex the eleven incident normals are orthonormal; direction i omits normal n_i. Cusps are future null rays, and their links are Euclidean ten-boxes. Two-faces have angles $\pi/2$ at finite vertices and zero at cusps.

Besides the two published dimension bounds, the external inputs are the standard acute-wall/link facts, simple cusp truncation, Dehn–Sommerville symmetry, and the published lower bound of 82 on the number of facets of a finite-volume right-angled eight-polytope [3, Sections 2–6]. The compact ceiling four makes every eight-face noncompact, so that lower bound applies. No arithmetic or facet-spectrum hypothesis is used.

2 Actual oppositions and a fixed null frame

Lemma 2.1. *Distinct unit outward normals have product zero or at most* -1.

Proof. Suppose their product g lies in $(-1, 0)$. Their Gram matrix $G = \left(\begin{smallmatrix} 1 & g \\ g & 1 \end{smallmatrix}\right)$ is positive definite and its inverse is entrywise nonnegative. For a strict future interior point x, let $b = (\langle x, u\rangle, \langle x, v\rangle)^T < 0$ and $x_0 = x - [u\ v]G^{-1}b$. The coefficients added to x are positive. Acuteness therefore makes all other walls strict at x_0. Also $x_0^2 = x^2 - b^T G^{-1} b < 0$ and $\langle x_0, x\rangle < 0$. Thus these walls meet along an actual ridge with a nonright angle, a contradiction. The same projection argument applies to a larger orthonormal set of actual walls. □

Call a triangular two-face with one finite and two ideal vertices a *bad triangle*. At q make a graph on its eleven directions, joining i, j when their polygon is a bad triangle. The ideal endpoints and the third wall are respectively

$$u_i = q - n_i, \qquad u_j = q - n_j, \qquad m_{ij} = q - n_i - n_j. \tag{1}$$

These formulas follow from the edge null equations, incidence, unit norm and the strict negative product of the third wall with q.

Lemma 2.2. *A triangle $S = \{i, j, k\}$ in this graph gives an actual three-face P_3, a triangular bipyramid with two finite and three ideal vertices. Its other finite vertex is*

$$q_S = 2q - n_i - n_j - n_k. \tag{2}$$

All six polygon facets of this three-face are bad triangles.

Proof. The three actual normals m_{ij}, m_{ik}, m_{jk} are orthonormal. The displayed q_S is future unit timelike, is perpendicular to these three normals and all n_l outside S, and has strictly negative products with n_i, n_j, n_k. For any remaining wall write its normal in the complete frame as $r = aq - \sum_l b_l n_l$, with $a > 0$ and $b_l \geq 0$. Acuteness with the three m walls gives $b_i + b_j, b_i + b_k, b_j + b_k \leq a$. Therefore $\langle r, q_S\rangle = -2a + b_i + b_j + b_k \leq -a/2 < 0$. This checks every wall, so q_S is an actual finite vertex.

In the Lorentz four-space of this three-face the six known walls are $y_l \geq 0$ and $y_l + y_m \leq t$ for $x = tq - \sum_{l\in S} y_l n_l$. They enclose exactly the finite-volume convex hull of q, q_S, u_i, u_j, u_k. The ideal points are feasible: they are the actual endpoints of the three edges at q. All five points lie in the closed chamber, so convexity prevents any other wall from cutting this hull. This proves the assertion for the whole three-face, not just its corner. □

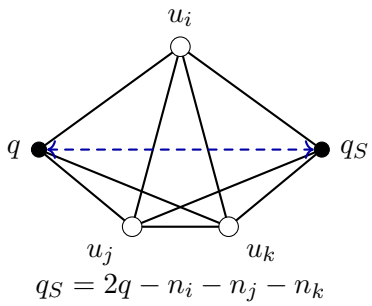


Figure 1: The actual P_3 three-face, drawn by incidence. Hollow vertices are ideal; the dashed opposition is not a polytope edge.

Let B count these three-faces. Each has two finite vertices and gives one undirected opposition between them. Conversely the graph triangle lemma identifies every opposition. Different triples at q give different opposites by linear independence of its normals. Thus the resulting graph on finite vertices has B edges. If $t(q)$ is its degree, then $\sum_q t(q) = 2B$.

Across an opposition the new normals are the eight old normals outside S and the three m normals. Their sum shows that

$$h = 3q - \sum_{i=1}^{11} n_i \quad \text{is invariant,} \qquad h^2 = 2, \quad \langle h, q\rangle = -3. \tag{3}$$

On each connected component fix one nonzero future null vector $Z \perp h$, possible because h is spacelike. The same Z is used at both ends of every actual opposition. Define

$$F(q) = -\langle Z, q\rangle > 0, \qquad p_i = -\langle Z, n_i\rangle / F(q). \tag{4}$$

Decomposition $Z = F(q)(q - \sum_i p_i n_i)$ gives exactly

$$\sum_{i=1}^{11} p_i = 3, \qquad \sum_{i=1}^{11} p_i^2 = 1. \tag{5}$$

Individual coordinates need not be positive. For any formal triple put $r_S = 2 - \sum_{i\in S} p_i$. Centering the eleven coordinates at $3/11$ and applying Cauchy to a centered triple indicator gives

$$\frac{13 - 4\sqrt{3}}{11} \le r_S \le \frac{13 + 4\sqrt{3}}{11}, \qquad \frac{11}{20} < r_S < \frac{20}{11}. \tag{6}$$

For an actual opposition, $r_S = F(q_S)/F(q)$ and the reverse ratio is r_S^{-1}. Formal triples are not asserted to be actual faces.

3 Three-face budgets

For a finite-volume right-angled three-polyhedron write F, s, p for its facets, cusps and finite vertices, and $\nu = F + s$. Finite and ideal degrees are three and four. Euler and polygon area give

$$p + 2s = 2F - 4, \quad p + 4s = 2\nu - 4, \qquad \sum_{\text{polygons}} (u + 2v - 5) = F + 2s - 12 \ge 0. \tag{7}$$

Here u, v are a polygon's finite and ideal vertex counts. Each summand is a nonnegative integer because its area is $\pi(u + 2v - 4)/2 > 0$.

Lemma 3.1. *One has $F \ge 6$ and $\nu \ge 9$. Equality in either $F = 6$ or $\nu = 9$ forces the whole polyhedron to be P_3.*

Proof. The area row also gives $p + 6s \ge 20$. If $F \le 5$, Euler gives $p + 2s \le 6$, $s \le 3$ and $p + 6s \le 18$, impossible. For $F = 6$, area gives $s \ge 3$. The total polygon corners are $24 - 2s$, so at least $2s$ of the six facets are triangles. Thus $s \le 3$, and $s = 3, p = 2$. All six facets are triangles, each having at most one finite vertex. Their six finite corners force all six to be bad. Lemma 2.2 identifies the polyhedron. Finally $F \ge 6$ and $F + 2s \ge 12$ imply $\nu \ge 9$, with equality only for $F = 6, s = 3$. □

Lemma 3.2. *Two bad triangular facets sharing an ideal–ideal edge force the whole three-polyhedron to be P_3.*

Proof. Normalize their shared future null endpoints u, v by their products -1 with the first finite apex. Its right angle gives $\langle u, v\rangle = -1$. Let n_1, n_2 be the two orthogonal triangle normals. The apices are $q_1 = u + v - n_2$ and $q_2 = tu + t^{-1}v - n_1$, $t > 0$: the right angle makes the product of the null coefficients one, and unit norm and orientation fix the normal coefficient -1. The remaining four wall normals are $u - n_2, v - n_2, tu - n_1, t^{-1}v - n_1$. Two cross products

are $-t, -t^{-1}$, so Lemma 2.1 forces $t = 1$. The point $w = u + v - n_1 - n_2$ is future null and feasible: it is the nonzero sum $(u - n_1) + (v - n_2)$ of actual outward normals; acuteness checks every other wall and strict interior pairing fixes its future orientation. The six walls enclose precisely the convex hull of q_1, q_2, u, v, w. Convexity gives the whole polyhedron. □

Lemma 3.3. *The noncanonical types with $\nu \leq 10$ are precisely the following count patterns; metric uniqueness is unnecessary:*

	F	s	p	A_0	A_1	A_2	A_3
Q_8	8	2	8	0	8	0	0
R_7	7	3	4	1	0	3	0

Here A_j counts finite vertices with exactly j ideal-ending edges.

Proof. At $\nu = 10$, area gives $F \leq 8$, and $F = 6$ is already canonical. For $F = 8, s = 2$, all polygons have zero excess. A bad triangle would have a zero-excess partner across its ideal edge, again a bad triangle, contrary to Lemma 3.2. There are no other triangles, and the total of 32 polygon corners makes every facet a quadrilateral with three finite and one ideal vertex. A finite vertex has at most one ideal-ending edge. The two cusps have no mutual edge, so their eight edges force exactly one at every finite vertex.

For $F = 7, s = 3, p = 4$, the total excess is one and there are 24 polygon corners. Hence at least four facets are triangles. Each ideal–ideal edge has at most one bad triangular facet, and there are at most three such edges, so at most three bad triangles occur. An all-ideal triangle uses the one excess unit. It follows that there are exactly three bad triangles, one ideal triangle, and three quadrilaterals of type $(u, v) = (3, 1)$. The ideal triangle uses all three ideal–ideal edges, leaving six finite–ideal and three finite–finite edges. No finite vertex has three ideal-ending edges: its three incident polygons would be bad triangles, giving P_3. The finite-edge graph on four vertices has three edges and no isolated vertex. If it were a triangle and isolated vertex the latter would have ideal degree three. It is therefore a tree. A path would have ideal degrees $2, 1, 1, 2$ and could support at most two bad triangles. The tree must be a three-leaf star, with ideal degrees $0, 2, 2, 2$. □

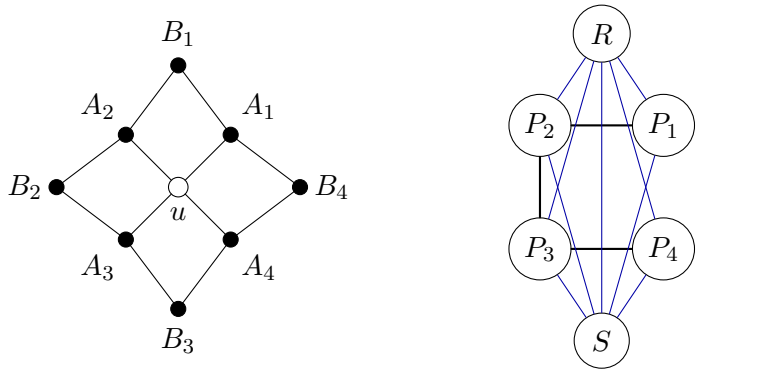
Lemma 3.4. *A right-angled three-polyhedron with one cusp has at least eleven facets.*

Proof. Equation (7) gives $F \geq 10$. If $F = 10$, every polygon has zero excess. The four facets at the cusp are quadrilaterals; the other six are finite pentagons. Write the four quadrilaterals cyclically as $Q_i = (u, A_i, B_i, A_{i+1})$, indices modulo four. Opposite Q_i have asymptotic planes meeting only at u, and each B_i belongs only to Q_i among the four. Thus the eight finite vertices displayed are distinct. Let P_i be the third facet at A_i. Its boundary contains $B_{i-1}A_iB_i$. At B_i, the three facets are Q_i, P_i, P_{i+1}, so consecutive P_i are distinct and adjacent. If $P_i = P_{i+2}$, its convex intersection with P_{i+1} contains both B_i, B_{i+1} and is the edge joining them. The boundary of P_{i+1} would then be the triangle $B_iA_{i+1}B_{i+1}$, impossible for a pentagon. Hence all four P_i are distinct and form four distinct pentagon adjacencies.

The eight quadrilateral–pentagon edges adjoin these four pentagons. The remaining two pentagons R, S adjoin no quadrilateral. Each adjoins all five other pentagons, since two facets share at most one edge. They give nine pentagon–pentagon edges, in addition to the four among the P_i. But the pentagon degree count gives exactly $(6 \cdot 5 - 8)/2 = 11$, contradicting at least thirteen. □

Call a three-face exceptional unless it is P_3 or Q_8. Define the nonnegative excess and signed defect

$$E = \sum_{Y \neq P_3} (\nu(Y) - 10), \qquad R = \sum_{Y \neq P_3} (8 - F(Y)). \tag{8}$$



four cusp quadrilaterals
eight distinct
boundary vertices

thirteen forced adjacencies
but the degree
count gives eleven

Figure 2: The one-cusp equality obstruction. The right panel is a forced facet-adjacency graph, not a planar embedding or a chamber realization.

Let X count exceptional finite-vertex/three-face incidences. Then

$$X \le 18E + 4R. \tag{9}$$

Indeed the right side of $p \le 18(\nu - 10) + 4(8 - F)$ minus p is $20\nu - 144 - 8F$. At $\nu = 10$ the exceptional R_7 has equality. At $\nu = 11$, area gives $F \le 10$ and Lemma 3.4 excludes $F = 10$, leaving a positive difference. For $\nu \ge 12$, use $F \le \nu$. Both budgets vanish on Q_8.

Lemma 3.5. *With sums over actual three-faces, put* $D = \sum_Y (A_2(Y) - 3A_0(Y))$ *and* $H = \sum_{Y \ne P_3} A_3(Y)$. *Then*

$$D \ge -30E, \qquad H \le 2E. \tag{10}$$

Proof. All three small types have $A_2 - 3A_0 = 0$. At $\nu = 11$, Lemma 3.4 and area give $F \le 9$, hence $p \le 10$. For $\nu \ge 12$, $p \le 2\nu - 4 \le 10(\nu - 10)$. Each remaining contribution to D is at least $-3p$, proving its bound.

For the second bound, first take an all-ideal-edge finite corner in a noncanonical three-face. Among its three incident polygons, every one that is not a bad triangle has excess at least one. Across the ideal edge of each bad triangle is another polygon. It is not a bad triangle by Lemma 3.2, so it too has excess at least one. These opposite polygons are distinct and avoid the chosen corner: their normals are the distinct $q - n_i - n_j$. There are thus three distinct excess units, giving $F + 2s \ge 15$. If $\nu \le 10$, then $s \ge 15 - \nu$ and $4s \le 2\nu - 5$ are incompatible. At $\nu = 11$ they force $s = 4, p = 2$, so $A_3 \le 2$. At $\nu = 12$, $A_3 \ge 5$ would give $p \ge 5$ and $4s \ge 3A_3 \ge 15$, hence $p + 4s \ge 21 > 20$ by integrality. Thus $A_3 \le 4$. Finally $4A_3 \le p + 4s = 2\nu - 4$ gives $A_3 \le (\nu - 2)/2 \le 2(\nu - 10)$ for $\nu \ge 13$. Summation proves $H \le 2E$ without a further metric classification. □

4 The strict ordinary incidence lower bound

At an ambient finite vertex let k edges end ideally and put $m = 11 - k$. Its contributions to D, H are $\binom{k}{2}m - 3\binom{m}{3}$ and $\binom{k}{3} - t(q)$. Every triple with zero or two ideal-ending directions is exceptional, so $X(q) \ge \binom{m}{3} + \binom{k}{2}m$. The integer row

$$40\binom{m}{3} + 28\binom{k}{2}m - 16\left(165 - \binom{k}{3}\right) \ge 0$$

has successive values $3960, 2160, 972, 288, 0, 0, 180, 432, 648, 720, 540, 0$ for $0 \le k \le 11$. It follows pointwise and then by summation that

$$31X \ge 16(165f - 2B) + 3D - 16H \ge 16(165f - 2B) - 122E. \tag{11}$$

Together with (9), this gives $680E + 124R \ge 2640f - 32B$.

Every polygon has nine three-cofaces, a finite vertex has 165, and a cusp has $2^8\binom{10}{2} = 11520$. Summing $F = 2 + p/2 + s$ and the definitions (8) gives

$$\begin{aligned} a_3 &= \frac{9}{2}a_2 - \frac{165}{4}f - 5760c, \\ E &= B - 10a_3 + 9a_2 + 11520c, \qquad R = 8a_3 - 9a_2 - 2B. \end{aligned} \tag{12}$$

The strict weighted row needed below is

$$a_2 > \frac{25}{2}f + 1930c. \tag{13}$$

Here is its direct Dehn–Sommerville derivation. Simple truncation has counts $b_0 = f + 1024c$, $b_{10} = a_{10} + c$ and $b_k = a_k + 2^{10-k}\binom{10}{k}c$ for $1 \le k \le 9$. Symmetry of $(t-1)^{11} + \sum_{k=0}^{10} b_k(t-1)^k$ gives

$$\begin{aligned} f &= 2a_6 - 12a_7 + 42a_8 - 112a_9 + 252a_{10} - 252c - 504, \\ a_2 &= 25a_6 - 148a_7 + 515a_8 - 1370a_9 + 3080a_{10} - 1220c - 6160. \end{aligned}$$

For $\lambda = 2123/1280$ these imply

$$55f + 5120\lambda c - \frac{22}{5}a_2 = -\frac{44}{5}a_7 + 44a_8 - 132a_9 + 308a_{10} - 616 < 0.$$

Indeed $4a_7 \ge 82a_8$ by the eight-face facet lower bound and $2a_9 \ge 11a_{10}$ by the elementary ten-face facet minimum. Both grouped pairs are nonpositive and the constant is strictly negative. This proves (13).

Eliminating a_2, a_3 in (12) gives

$$R = -\frac{5}{4}B - \frac{165}{8}f + 5760c - \frac{3}{4}E, \qquad 360c < B - \frac{75}{2}f - E.$$

Consequently $680E + 124R \ge 2640f - 32B$ becomes $587E - 123B - 10395f/2 + 714240c \ge 0$. Since $714240 = 1984 \cdot 360$, its left side is strictly less than $1861B - 159195f/2 - 1397E$. We have proved

$$\boxed{3722B > 159195f + 2794E.} \tag{14}$$

5 A structured scalar estimate

The analytic input is proved by the subset cases below.

Proposition 5.1. *Let eleven real numbers satisfy* (5)*, and let I be any k-element subset. For $r_S = 2 - \sum_{i \in S} p_i$,*

$$\sum_{\substack{S \subset I \\ |S|=3}} (r_S^{-2} - \frac{37}{100})_+ \le \frac{756}{25} + \frac{27}{100}\binom{k}{3}. \tag{15}$$

Write $\lambda = 37/100$ and $\rho = 10/\sqrt{37}$. For $3 \le k < 11$, set

$$S = \sum_I p_i, \quad y_i = p_i - S/k, \quad V = \sum_I y_i^2, \quad \mu = 2 - 3S/k, \quad d_T = -\sum_{i \in T} y_i.$$

Cauchy on the complement gives

$$0 \le V \le v_k(S) := 1 - S^2/k - (3-S)^2/(11-k), \quad \frac{3k - \sqrt{2k(11-k)}}{11} \le S \le \frac{3k + \sqrt{2k(11-k)}}{11}. \tag{16}$$

In particular $V \le 2/11$. Uniform averages over the $\binom{k}{3}$ triples satisfy, with $T_j = \sum_I y_i^j$,

$$\begin{aligned} &\mathbb{E}d = 0, && \mathbb{E}d^2 = \alpha_2 V, && \mathbb{E}d^3 = -\alpha_3 T_3, \quad \mathbb{E}d^4 = \alpha_4 T_4 + b_4 V^2, \\ &\alpha_2 = \frac{3(k-3)}{k(k-1)}, && \alpha_3 = \frac{3(k-3)(k-6)}{k(k-1)(k-2)}, \\ &\alpha_4 = \frac{3(k^2-17k+54)}{k(k-1)(k-2)}, && b_4 = \frac{18(k-4)}{k(k-1)(k-2)}. \end{aligned} \tag{17}$$

These follow by expanding products of indicators: the inclusion probabilities of one, two and three specified indices are $3/k$, $6/[k(k-1)]$ and $6/[k(k-1)(k-2)]$. The fourth expansion uses $\sum_{i \ne j} y_i^3 y_j = -T_4$, $\sum_{i<j} y_i^2 y_j^2 = (V^2 - T_4)/2$, and $\sum_i y_i^2 \sum_{j<l,\ j,l \ne i} y_j y_l = T_4 - V^2/2$.

Two useful moment bounds are

$$T_4 \ge V^2/k + T_3^2/V, \qquad |T_3| \le \frac{k-2}{\sqrt{k(k-1)}} V^{3/2}. \tag{18}$$

The first is the orthogonal projection of (y_i^2) onto the vectors 1 and y; interpret it by continuity at $V = 0$. For the second, $y_i \le M = \sqrt{(k-1)V/k}$. Put $b = -M/(k-1)$. Expanding $\sum_i (M - y_i)(y_i - b)^2 \ge 0$ bounds T_3 from above by $(k-2)MV/(k-1)$; replace y by $-y$ for the lower bound.

For $r = \mu + d > 0$ the exact identity

$$r^{-2} = \mu^{-2} - 2d\mu^{-3} + 3d^2\mu^{-4} - 4d^3\mu^{-5} + \frac{4r + \mu}{\mu^5 r^2} d^4 \tag{19}$$

follows by clearing denominators. The remainder factor decreases in r. With $a = 11/20$, it is at most $K(\mu) = (4a + \mu)/(\mu^5 a^2)$. The positive part is recovered by

$$\mathbb{E}(r^{-2} - \lambda)_+ = \mathbb{E}r^{-2} - \lambda + \mathbb{E}(\lambda - r^{-2})_+. \tag{20}$$

The last correction is at most $27/400$ by (6). If $b = \rho - \mu > 0$, concavity of $\lambda - r^{-2}$ above ρ and $(d-b)_+ \le d^2/(4b)$ give

$$\mathbb{E}(\lambda - r^{-2})_+ \le \frac{\lambda^{3/2} \mathbb{E}d^2}{2(\rho - \mu)}. \tag{21}$$

Here $\lambda^{3/2} < 113/500$ and $\rho > 18/11$; the correction vanishes when an upper bound on r is at most $18/11$.

For $k \le 5$, each summand in (15) is less than $121/36 - 37/100 = 673/225$, since $r > 6/11$. There are at most ten triples, giving a total below $756/25$. For $k = 6$, pair every triple with its complementary triple in I. Their ratios sum to $4 - S > 18/11$, by (16) and $S \le (18 + 2\sqrt{15})/11 < 26/11$. The function $(r^{-2} - \lambda)_+$ is convex and decreasing. With both ratios $> 6/11$, each pair is bounded by its values at $6/11$ and $12/11$, namely $12461/3600$ in total. There are ten pairs; $12461/360 < 891/25$, the required right side. Empty triple sums need no separate interpretation. This proves all cases $k \le 6$.

6 Seven, eight and eleven directions

For $k = 7, 8$, the coefficients in (17) have $\alpha_3 > 0$, $\alpha_4 < 0$ and $b_4 + \alpha_4/k > 0$. Use $|T_3| \le V^{3/2}$ and $T_4 \ge V^2/k$ in (19). The average reciprocal is at most

$$J_k(\mu, V) = \mu^{-2} + 3\alpha_2 V\mu^{-4} + 4\alpha_3 V^{3/2}\mu^{-5} + K(\mu)(b_4 + \alpha_4/k)V^2. \tag{22}$$

Every displayed term is nonnegative, increases with V and decreases with positive μ. For clarity, we give the full rational interval certificate contract used to bound this expression; no sampled frames or floating-point inequalities are required.

The admissible S intervals in (16) are contained in $[27/22, 57/22]$ for $k = 7$ and $[17/11, 31/11]$ for $k = 8$. Split these enclosing intervals into closed intervals $[j/1100, (j+1)/1100]$, with respectively $1350 \le j < 2850$ and $1700 \le j < 3100$. For each interval $[l, u]$ set

$$S_* = \min(u, \max(l, 3k/11)), \quad V_* = v_k(S_*), \quad \mu_- = 2 - 3u/k, \quad \mu_+ = 2 - 3l/k.$$

This is the exact maximum of the concave quadratic v_k on the interval. If $V_* < 0$ the interval is empty of admissible frames. Otherwise evaluate (22) at (μ_-, V_*), rounding $\sqrt{V_*}$ upward to a multiple of 10^{-6}. The rounding is certified by integer squaring, not numerical tolerance. Add the least of the following applicable correction bounds:

$$\frac{27}{400}, \qquad \frac{(113/500)\alpha_2 V_*}{2(18/11 - \mu_+)} \quad (\mu_+ < 18/11), \qquad 0 \quad \left(\mu_+ + \sqrt{3(k-3)V_*/k} \le 18/11\right).$$

For the last test the square root is again rounded upward. These bounds follow from (21) and the centered triple indicator norm. The resulting rational number is strictly less than

$$\lambda + \frac{756/25 + (27/100)\binom{k}{3}}{\binom{k}{3}} \tag{23}$$

on every nonempty interval. The exact certificate records these checks and their least gaps:

k	enclosing intervals	nonempty	empty	least gap greater than
7	1500	1498	2	6/25
8	1400	1386	14	3/50

The intervals cover the whole domain and the envelopes bound each whole interval. Thus these finite rational inequalities prove (15) for $k = 7, 8$.

For $k = 11$, $\mu = 13/11$, $V = 2/11$ and

$$\mathbb{E}d^2 = 24/605, \quad \mathbb{E}d^3 = -4T_3/33, \quad \mathbb{E}d^4 = 7V^2/55 - 2T_4/55.$$

Here $K(13/11) < 5$, so first replace the whole nonnegative fourth moment remainder by five times that moment. Then use the joint bound in (18). The reciprocal average is at most

$$\mu^{-2} + \frac{3(24/605)}{\mu^4} + 5\frac{15}{121}V^2 + \frac{16T_3}{33\mu^5} - T_3^2.$$

The sharp cubic bound gives $|T_3| \le 18/(121\sqrt{5}) < 1/15$. For $C = 16/(33\mu^5) > 2/15$, the function $CT_3 - T_3^2$ is increasing on $[-1/15, 1/15]$, so it is at most $C/15 - 1/225$. Equation (21) bounds the correction by $(113/500)(24/605)/(2(5/11))$. The resulting positive-part mean is

$$\frac{273313374321407}{611561341462500} < \frac{2493}{5500} = \frac{756/25 + (27/100)\cdot 165}{165}.$$

This proves the full-subset case by rational comparisons alone.

7 Nine directions: a global analytic bound

For a nine-element set I, write $\delta = 3 - S$. Equation (16) gives

$$0 \le \delta \le 12/11, \quad \mu = 1 + \delta/3, \quad 0 \le V \le 2\delta/3 - 11\delta^2/18 \le 2/11.$$

We prove the stronger mean estimate

$$\mathbb{E}(r^{-2} - \lambda)_+ \le \frac{63}{100} - \frac{\delta}{18}. \tag{24}$$

The moments here are $\mathbb{E}d^2 = V/4$, $\mathbb{E}d^3 = -3T_3/28$ and $\mathbb{E}d^4 = 5V^2/28 - 3T_4/28 \le V^2/6$. The cubic bound (18) gives

$$\frac{3}{7}T_3 \le \frac{V^{3/2}}{2\sqrt{2}} \le \frac{V}{2\sqrt{11}} \le \frac{V}{6}.$$

For the clipping correction, $\rho > 18/11$ and $\mu \le 15/11$ imply $b = \rho - \mu > 3/11$. The elementary global inequality

$$(d - b)_+ \le \frac{27d^4}{256b^3}$$

follows by maximizing $(d - b)/d^4$ for $d > b$, at $d = 4b/3$. Using the tangent bound as in (21) gives

$$(\lambda - r^{-2})_+ \le \frac{54d^4}{256\rho^3 b^3} \le \frac{11^6}{128 \cdot 18^3} d^4 \le 3d^4.$$

In (19), $\mu \ge 1$, $r > 11/20$ give a remainder coefficient at most $r^{-2} + 4r^{-1} < 1280/121$. Combining these estimates, with their signs unchanged, yields

$$\mathbb{E}(r^{-2} - \lambda)_+ + \lambda \le \mu^{-2} + \frac{3V}{4\mu^4} + \frac{V}{6\mu^5} + \frac{1643}{726}V^2. \tag{25}$$

Put $x = \delta/3$, so $0 \le x \le 4/11$ and $V \le v = x(4 - 11x)/2$. The right side increases with V. After substituting v, the exact cleared difference is

$$(1+x)^5 \left[1 - \frac{x}{6} - \left((1+x)^{-2} + \frac{3v}{4(1+x)^4} + \frac{v}{6(1+x)^5} + \frac{1643}{726}v^2\right)\right] = x^2 R(x),$$

where

$$R(x) = \frac{233}{24} + \frac{275}{24}x + \frac{10}{3}x^2 + \frac{x^3}{6} - \frac{x^4}{6} - \frac{1643}{2904}(4 - 11x)^2(1+x)^5.$$

The first five terms are at least $233/24$ since $0 \le x < 1$. The derivative of the last product without its coefficient is $(4 - 11x)(1+x)^4(-2 - 77x) \le 0$, so the product is at most 16. Thus $R(x) \ge 233/24 - 1643 \cdot 16/2904 = 635/968 > 0$. This proves (24); multiplying by 84 gives $1323/25 = 756/25 + (27/100)\binom{9}{3}$, as required. Equality is possible only at $\delta = 0$, when the nine coordinates are $1/3$ and the other two zero. No interval certificate is needed for this case.

8 Ten directions: joint moments and rational polynomials

Let v be the omitted coordinate. The remaining ten have

$$\frac{3 - 2\sqrt{5}}{11} \le v \le \frac{3 + 2\sqrt{5}}{11}, \quad \mu = \frac{11 + 3v}{10}, \quad V = \frac{1 + 6v - 11v^2}{10}. \tag{26}$$

Their moments are $\mathbb{E}d^2 = 7V/30$, $\mathbb{E}d^3 = -7T_3/60$ and $\mathbb{E}d^4 = 3V^2/20 - T_4/15$. Set $w = \sqrt{21V/10}$, $a = \mu - w$, and $\gamma = 8/\sqrt{90}$. The triple range and the sharp cubic bound give $a > 11/20$ and $|T_3| \le L = \gamma V^{3/2}$. The exact remainder and the joint fourth moment imply

$$\mathbb{E}r^{-2} \le \mu^{-2} + \frac{7V}{10\mu^4} + \max_{|T| \le L}\left\{\frac{7T}{15\mu^5} + K\left(\frac{43V^2}{300} - \frac{T^2}{15V}\right)\right\}, \quad K = \frac{4a + \mu}{\mu^5 a^2}. \tag{27}$$

At $V = 0$ the bracket is zero. On the actual cubic domain the coefficient of K is nonnegative, because $43/300 - \gamma^2/15 = 259/2700 > 0$. This observation is essential when increasing K; its positive and negative occurrences cannot be changed independently.

In fact $K < 7$. For $v \geq 0$, $\mu \geq 11/10$, $a > 11/20$ give $K \leq 12000000/1771561 < 7$. For $v \leq 0$, $V \leq 1/10$ and $V' \geq 3/5$, so $a' = 3/10 - (21/20)V'/\sqrt{21V/10} < 0$. Hence $a \geq a(0) = (11 - \sqrt{21})/10 > 16/25$. Also $v > -1/6$ gives $\mu > 21/20$. Consequently $K < 7220000000/1045529856 < 7$. The endpoint $V = 0$ follows by continuity. We may replace the whole K in (27) by seven.

For $v \leq 0$, the vertex $V/(2\mu^5)$ of the concave quadratic is beyond L, since $2\gamma\mu^5\sqrt{V} \leq (8/15)(11/10)^5 < 1$. Using $\gamma < 17/20$ and $\sqrt{V} \leq 1/10 + (5/2)V$ gives

$$J_- = \mu^{-2} + \frac{7V}{10\mu^4} + \frac{119V/3000 + 119V^2/120}{\mu^5} + \frac{1813}{2700}V^2. \tag{28}$$

The clipping correction is zero, since $\mu + w < 39/25 < 41/25 < \rho$. For $v \geq 0$, completing the square without constraining T gives

$$J_+ = \mu^{-2} + \frac{7V}{10\mu^4} + \frac{301}{300}V^2 + \frac{7V}{60\mu^{10}}. \tag{29}$$

For $0 \leq v \leq 1/20$, one has $\mu \leq 223/200$, $V \leq 509/4000$, $w < 13/25$, and hence $\mu + w < 327/200 < 41/25$; the correction is again zero. On the rest of the domain, (21), $\lambda^{3/2} < 113/500$ and $\rho > 41/25$ give the correction bound

$$C_+ \leq \frac{791V}{30000(41/25 - \mu)}. \tag{30}$$

The denominator stays positive on the larger interval $[1/20, 7/10]$.

The required positive-part mean is $261/500$, equivalently the corrected reciprocal mean is at most $223/250$. Clearing the positive denominators in (28)–(30) reduces this to the positivity of three explicit rational polynomials:

$$\begin{aligned}
P_- &= \frac{223}{250}\mu^5 - \mu^3 - \frac{7}{10}V\mu - \frac{119}{3000}V - \frac{119}{120}V^2 - \frac{1813}{2700}V^2\mu^5,\\
P_0 &= \frac{223}{250}\mu^{10} - \mu^8 - \frac{7}{10}V\mu^6 - \frac{301}{300}V^2\mu^{10} - \frac{7}{60}V,\\
P_c &= (41/25 - \mu)P_0 - \frac{791}{30000}V\mu^{10}.
\end{aligned} \tag{31}$$

Here μ, V are the polynomials in (26). The respective domains, all enclosing the appropriate actual domain, are $[-7/50, 0]$, $[0, 1/20]$ and $[1/20, 7/10]$. Some outer points have $V < 0$; only polynomial positivity, not a moment interpretation, is asserted at these extraneous points.

For completeness the exact finite positivity certificate is as follows. For a degree-m rational polynomial Q, expand $Q(l + (u - l)t) = \sum_{j=0}^{m} c_j t^j$. Its Bernstein coefficients are

$$\beta_i = \sum_{j=0}^{i} c_j \frac{\binom{i}{j}}{\binom{m}{j}}, \qquad Q(l + (u-l)t) = \sum_{i=0}^{m} \beta_i \binom{m}{i} t^i (1-t)^{m-i}. \tag{32}$$

The basis is nonnegative and sums to one on $[0, 1]$. On each consecutive interval of length $1/1000$, every coefficient computed from (31) and (32) is strictly positive. The complete table of obligation sizes is

polynomial	domain	intervals	positive coefficients
P_-, degree 9	$[-7/50, 0]$	140	1400
P_0, degree 14	$[0, 1/20]$	50	750
P_c, degree 15	$[1/20, 7/10]$	650	10400

Thus all 12550 coefficients prove positivity on all 840 closed intervals. They are recorded exactly, with digests and least coefficients, in the accompanying certificate; the independent replay recomputes them from the three displayed polynomials. This completes the ten-direction case, and hence Proposition 5.1.

9 Actual-subset deletion and the contradiction

At a finite vertex q, let I consist of the k directions whose edges end ideally. Its $t(q)$ actual P_3 triples are a subset of the $\binom{k}{3}$ formal triples in I. For each actual triple, $r^{-2}-1 \le (r^{-2}-37/100)_+ - 63/100$. Inserting other formal triples is legitimate because the added terms are nonnegative. Proposition 5.1 therefore gives

$$\sum_{\text{actual } S} (r_S^{-2}-1) \le \frac{9}{25}(84-t(q)) + \frac{27}{100}\left(\binom{k}{3} - t(q)\right). \tag{33}$$

The last parenthesis counts actual noncanonical three-faces whose finite corner has three ideal-ending edges. Summed over ordinary finite vertices it is exactly H, not an arbitrary graph error.

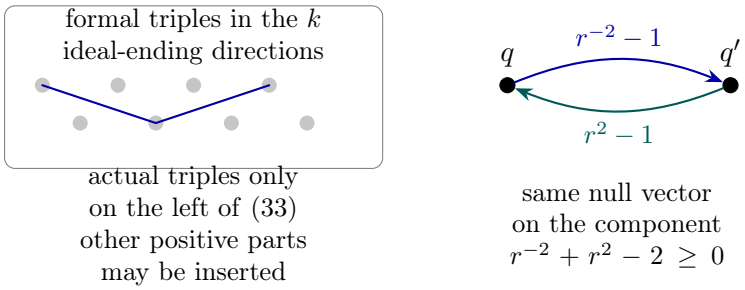


Figure 3: Deletion and reversal use different domains. The scalar bound enlarges an actual subset; the final sum pairs only actual opposition edges. Neither panel asserts that formal triples are feasible faces.

Sum (33) over all finite vertices, with the fixed null choice on each component. Every undirected opposition contributes $r^{-2}+r^2-2=(r-r^{-1})^2\ge 0$. As $\sum t = 2B$ and $H \le 2E$,

$$0 \le \frac{9}{25}(84f-2B) + \frac{27}{100}H \le \frac{9}{25}(84f-2B) + \frac{27}{50}E.$$

Consequently

$$\boxed{4B \le 168f + 3E.} \tag{34}$$

This is an ordinary unweighted finite-vertex sum; it does not use a maximum-height vertex or substitute a weighted average.

Twice (14) and 1861 times (34) say

$$7444B > 318390f + 5588E, \qquad 7444B \le 312648f + 5583E.$$

They would imply $5742f + 5E < 0$, contrary to $f > 0$, $E \ge 0$. There is no mixed right-angled eleven-polytope. The eleven-face argument in Section 1 proves Theorem 1.1.

Proof and verification record

The proof separates universal geometry, exact incidence algebra and finite scalar positivity. All local geometric arguments used by the dimension exclusion are included above. Their independent written reviews do not amount to external refereeing or full geometric formalization. The

only finite positivity supplements are the complete seven/eight-direction rational interval envelopes and the three ten-direction Bernstein polynomial obligations. They are interval proofs, not a census of polytopes, a numerical search or sampled frames.

The exact standard-library checkers and compact certificates are:

- `scripts/check_mixed_eleven_balanced_and_subset_rows_20261020.py`, with `results/mixed_eleven_balanced_and_subset_rows_20261020.json`;
- `scripts/check_mixed_eleven_nine_direction_global_20261020.py`, with `results/mixed_eleven_nine_direction_global_20261020.json`;
- `scripts/check_mixed_k10_rational_envelopes_20261020.py`, with `results/right_angled_mixed_k10_rational_envelopes_20261020.json`.

Each has passed generation and a separate read-only exact replay. The certificate archive stores the rational gaps, complete coefficient digests, source bindings and bounded execution ledgers. An older nine-direction interval certificate is preserved as independent support; the global proof in Section 7 is sufficient here.

The standard-library module `formal/CoxeterProofs/RightAngledMixed10.lean` has compiled in a single capped batch and passed a separate printed-axiom audit. Its primitive count identities, strict weighted row and three-face budgets are explicit premises. It derives the balanced lower bound; the summed real analytic transport inequality is a separate explicit premise. From those inputs it checks the upper bound and contradiction, and records the eleven-face dimensional tail as an interface. The eight audited theorems use only `propext` and `Quot.sound`; the integer comparison is axiom-free and there is no `sorryAx`. The actual-wall geometry and the scalar-to-frame interpretation remain written arguments, rather than hidden published axioms or claims of end-to-end Lean verification.

References

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[3] S. Alexandrov, *On ideal vertices of right-angled hyperbolic polyhedra*, 2023, Sections 2–6. `https://arxiv.org/html/2303.09533v1`.